

# DIFFERENTIAL EQUATIONS

## Lecture 5 — Exact Differential Equations

### The Total Differential

#### Definition: Total Differential

For a function  $u(x, y)$ , the **total differential** is:

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

If  $u(x, y) = C$  (a constant), then  $du = 0$ , giving:

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 \quad \Longleftrightarrow \quad M(x, y) dx + N(x, y) dy = 0$$

where we identify  $M = \partial u / \partial x$  and  $N = \partial u / \partial y$ .

### Second-Order Partial Derivatives

For a function  $u(x, y)$ , we can differentiate partially twice in four ways:

$$u_{xx} = \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial x} \right), \quad u_{xy} = \frac{\partial}{\partial y} \left( \frac{\partial u}{\partial x} \right), \quad u_{yx} = \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial y} \right), \quad u_{yy} = \frac{\partial}{\partial y} \left( \frac{\partial u}{\partial y} \right)$$

The two **mixed partial derivatives**  $u_{xy}$  and  $u_{yx}$  are the key ones here.

#### Clairaut's Theorem

If  $u(x, y)$  and its partial derivatives are **continuous**, then the mixed partials are equal:

$$u_{xy} = u_{yx} \quad \text{i.e.,} \quad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

Since  $M = \partial u / \partial x$  and  $N = \partial u / \partial y$ , this gives us:

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

This equality is both the test for exactness and the reason the solution method works.

### Exact ODEs — Definition and Test

#### Definition: Exact ODE

The equation  $M(x, y) dx + N(x, y) dy = 0$  is called **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

## Solution Method

### Solving an Exact ODE

Once exactness is confirmed, the general solution is found using:

$$\int M(x, y) dx + \int [\text{terms of } N \text{ that are free from } x] dy = C$$

#### Steps:

1. Identify  $M$  and  $N$ .
2. Check  $\partial M/\partial y = \partial N/\partial x$ . If equal, the equation is exact.
3. Integrate  $M$  with respect to  $x$  (treating  $y$  as a constant throughout).
4. Look at  $N$  and pick out **only** the terms that contain **no**  $x$  at all. Integrate those w.r.t.  $y$ .
5. Add the results from steps 3 and 4, set equal to  $C$ .

**Why only the free-from- $x$  terms of  $N$ ?** Integrating  $M$  w.r.t.  $x$  already captures every term that involves  $x$ . The only terms we could possibly be missing are the ones in  $N$  that have *no*  $x$  in them — those are invisible to the  $\int M dx$  step. So we only add those, to avoid counting anything twice.

## What “Free from $x$ ” Means

Go through  $N$  term by term. For each term, ask: **does this term involve  $x$  in any way?**

| Term in $N$  | Contains $x$ ? | What to do                               |
|--------------|----------------|------------------------------------------|
| $x^2$        | Yes            | Skip — already covered by $\int M dx$    |
| $3xy$        | Yes            | Skip                                     |
| $e^x \cos y$ | Yes            | Skip                                     |
| $\sin x$     | Yes            | Skip                                     |
| $5y^2$       | No             | Include: $\int 5y^2 dy = \frac{5y^3}{3}$ |
| $\ln y$      | No             | Include: $\int \ln y dy$                 |
| 3 (constant) | No             | Include: $\int 3 dy = 3y$                |
| $y \sin y$   | No             | Include: $\int y \sin y dy$              |

If **no** terms in  $N$  are free from  $x$ , then  $\int N_{\text{free}} dy = 0$  and the solution is simply  $\int M dx = C$ .

## Worked Examples

**Example 1:**  $(y + 2xy^2) dx + (x + 2x^2y) dy = 0$ 

$$M = y + 2xy^2, \quad N = x + 2x^2y$$

**Test:**

$$\frac{\partial M}{\partial y} = 1 + 4xy \quad \frac{\partial N}{\partial x} = 1 + 4xy \quad \checkmark \text{ EXACT}$$

**Integrate  $M$  w.r.t.  $x$ :**

$$\int (y + 2xy^2) dx = xy + x^2y^2$$

**Terms of  $N$  free from  $x$ :**  $N = x + 2x^2y$  — both terms contain  $x$ . Nothing to add.

$$\boxed{xy + x^2y^2 = C}$$

**Example 2:**  $2xy dx + x^2 dy = 0$ 

$$M = 2xy, \quad N = x^2$$

**Test:**

$$\frac{\partial M}{\partial y} = 2x \quad \frac{\partial N}{\partial x} = 2x \quad \checkmark \text{ EXACT}$$

**Integrate  $M$  w.r.t.  $x$ :**

$$\int 2xy dx = x^2y$$

**Terms of  $N$  free from  $x$ :**  $N = x^2$  contains  $x$ . Nothing to add.

$$\boxed{x^2y = C}$$

**Example 3:**  $(2xy + y \cos x) dx + (x^2 + \sin x + 3y^2) dy = 0$ 

$$M = 2xy + y \cos x, \quad N = x^2 + \sin x + 3y^2$$

**Test:**

$$\frac{\partial M}{\partial y} = 2x + \cos x \quad \frac{\partial N}{\partial x} = 2x + \cos x \quad \checkmark \text{ EXACT}$$

**Integrate  $M$  w.r.t.  $x$ :**

$$\int (2xy + y \cos x) dx = x^2y + y \sin x$$

**Terms of  $N$  free from  $x$ :** In  $N = x^2 + \sin x + 3y^2$ , the terms  $x^2$  and  $\sin x$  contain  $x$ . The term  $3y^2$  is free. Integrate it:

$$\int 3y^2 dy = y^3$$

$$\boxed{x^2y + y \sin x + y^3 = C}$$

**Example 4:**  $(3x^2y - 6x) dx + (x^3 + 2y) dy = 0$ 

$$M = 3x^2y - 6x, \quad N = x^3 + 2y$$

**Test:**

$$\frac{\partial M}{\partial y} = 3x^2 \quad \frac{\partial N}{\partial x} = 3x^2 \quad \checkmark \text{ EXACT}$$

**Integrate  $M$  w.r.t.  $x$ :**

$$\int (3x^2y - 6x) dx = x^3y - 3x^2$$

**Terms of  $N$  free from  $x$ :** In  $N = x^3 + 2y$ , the term  $x^3$  contains  $x$ . The term  $2y$  is free. Integrate it:

$$\int 2y dy = y^2$$

$$\boxed{x^3y - 3x^2 + y^2 = C}$$

**Example 5:**  $(e^x \sin y - 2y \sin x) dx + (e^x \cos y + 2 \cos x + y^2) dy = 0$ 

$$M = e^x \sin y - 2y \sin x, \quad N = e^x \cos y + 2 \cos x + y^2$$

**Test:**

$$\frac{\partial M}{\partial y} = e^x \cos y - 2 \sin x \quad \frac{\partial N}{\partial x} = e^x \cos y - 2 \sin x \quad \checkmark \text{ EXACT}$$

**Integrate  $M$  w.r.t.  $x$ :**

$$\int (e^x \sin y - 2y \sin x) dx = e^x \sin y + 2y \cos x$$

**Terms of  $N$  free from  $x$ :** In  $N = e^x \cos y + 2 \cos x + y^2$ , both  $e^x \cos y$  and  $2 \cos x$  contain  $x$ . The term  $y^2$  is free. Integrate it:

$$\int y^2 dy = \frac{y^3}{3}$$

$$\boxed{e^x \sin y + 2y \cos x + \frac{y^3}{3} = C}$$

## Summary

| Concept            | Key Point                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------|
| Total differential | $du = \partial u / \partial x dx + \partial u / \partial y dy$ ; if $u = C$ then $du = 0$ |
| Clairaut's theorem | Mixed partials are equal: $u_{xy} = u_{yx}$                                               |
| Exactness test     | Check $\partial M / \partial y = \partial N / \partial x$                                 |
| Solution formula   | $\int M dx + \int [\text{terms of } N \text{ free from } x] dy = C$                       |
| Free from $x$      | Term in $N$ with no $x$ in it whatsoever — integrate w.r.t. $y$                           |
| No free terms      | If all terms of $N$ contain $x$ , solution is just $\int M dx = C$                        |

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*End of Lecture 5*